

TSA SERIES

MODEL

TSA700-6N

The TSA700-6N is a horizontal-multi-joint SCARA robot built for repetitive planar work — assembly, part loading, sorting, inspection and dispensing. 6 kg payload, 700 mm arm reach, 200 mm Z stroke and a fast 0.44 s standard cycle, in a compact 24 kg body. Integrated pneumatic and signal harness through the arm keeps end-effector tooling clean and repeatable.

4-Axis SCARA

Payload 6 kg · Reach 700 mm

Repeatability

±0.02 mm (XY) positional accuracy

Texsonics RC

In-house controller, drives and teach pendant

Site-ready

Floor / Wall / Ceiling mount, factory calibrated

01 SPECIFICATIONS

MECHANICAL

AXES	4 (SCARA)
PAYLOAD	6 kg
ARM LENGTH	700 mm
Z STROKE	200 mm
STANDARD CYCLE	0.44 s
BODY WEIGHT	24 kg

ENVIRONMENT / CONTROL

SIGNALS	15 pin D-Sub (13-return cable)
PNEUMATICS	1 × Ø6 + 2 × Ø6 (0.59 MPa)
HARNESS LENGTH	3 m
PROTECTION	IP20
ENVIRONMENT	0–40 °C, 35–85 % RH

02 PRODUCT



TSA700-6N · TSA SERIES

03 WORKING RANGE

MOTION — GROUP A	
A-AXIS (INNER ARM)	±140°
B-AXIS (OUTER ARM)	±145°
Z-AXIS (VERTICAL STROKE)	200 mm
W-AXIS (TOOL ROTATION)	±360°
A/B MAX SPEED	8700 mm/s

MOTION — GROUP B	
Z MAX SPEED	1100 mm/s
W MAX SPEED	2000°/s
REPEATABILITY — A/B	±0.02 mm
REPEATABILITY — Z	±0.01 mm
REPEATABILITY — W	±0.01°

04 OVERVIEW

HIGH-THROUGHPUT PLANAR MOTION, 0.44 S CYCLE.

The TSA700-6N is a 4-axis SCARA built for repetitive planar work — assembly, part loading, sorting, inspection and dispensing. Integrated pneumatic and signal harness keeps end-effector tooling clean and repeatable.

01 **FAST CYCLE**
0.44 s standard cycle, 8700 mm/s XY speed and 2000°/s tool rotation for the shortest possible takt times.

02 **HARNESS-THROUGH ARM**
15-pin D-Sub signal and dual 6 mm pneumatic lines routed through the arm keep tooling wiring clean.

03 **PLANAR ACCURACY**
±0.02 mm XY, ±0.01 mm Z, ±0.01° tool — precision suited to assembly and inspection cells.

05 CONTROL & INTEGRATION

CONTROLLER
Texsonics RC series
in-house motion controller, dual-kernel real-time OS

TEACH PENDANT
11.5" touch tablet
drag-teaching, Wi-Fi, on-device offline programming

FIELDBUS
EtherCAT / Modbus TCP
industrial fieldbus, dual-protocol support

SOFTWARE
TEXCAM offline CAM
toolpath generation, digital twin simulation, vision

06 APPLICATIONS

Deployable across the primary applications the TSA700-6N was engineered for — full production-line integration with the Texsonics control stack.

Assembly

Loading / Unloading

Sorting

Inspection

Dispensing

Machine Tending

CNC Loading

Palletizing

Packaging

Vision Inspection

07

BUILT FOR THE OPERATOR

- Live 3D position display
- Online command execution
- Position editor
- Variable monitor
- I/O monitor
- Error diagnostics
- Dynamic teach-in
- Digital twin simulation

08

SERVICE & SUPPORT**Made in Coimbatore.
Supported from
Coimbatore.**

Every TSA700-6N is engineered, built and calibrated in our Coimbatore facility. Field service, spares and applications engineering ship from the same address — no import lead times, no time-zone escalations.

STANDARD WARRANTY

12 months on the mechanical assembly, 12 months on the controller.

APPLICATIONS ENGINEERING

Cell design, cycle-time studies and end-of-arm tooling by our in-house team, included on every project.

SPARES & FIELD SERVICE

Direct-to-plant spares from Coimbatore, with pan-India field service engineers on 24-hour dispatch.

09

READY TO INTEGRATE**Talk to an engineer about
integrating the TSA700-6N
into your line.**

Applications engineering, cycle-time sizing, and turnkey cell design are included on every project.

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TSA700-6N

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TSA SERIES

4-AXIS SCARA • PAYLOAD 6 kg • REACH 700 mm

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